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What is a Network?



- In its simplest form, a network is nothing more than "two connected computers sharing resources with one another."

It is composed of two main aspects:

- **Physical Connection** (wires, cables, wireless media)
- **Logical Connection** (data transporting across the physical media)

Why Build a Computer Network?

- Before computer networks, people sent and received information by hand, using the postal service. This is slow and can be unreliable.
- Computer networks enable faster, more efficient modes of communication, i.e., email, video conferencing, etc.
- Computer networks and the sharing of electronic data encourage the use of standard policies and procedures.
- Computer networks provide backup and recovery support for our data, i.e., redundancy.
- Computer networks lead to cost savings.

Types of Computer Networks (by Size)

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Wireless Local Area Network (WLAN)
- Campus Area Network (CAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

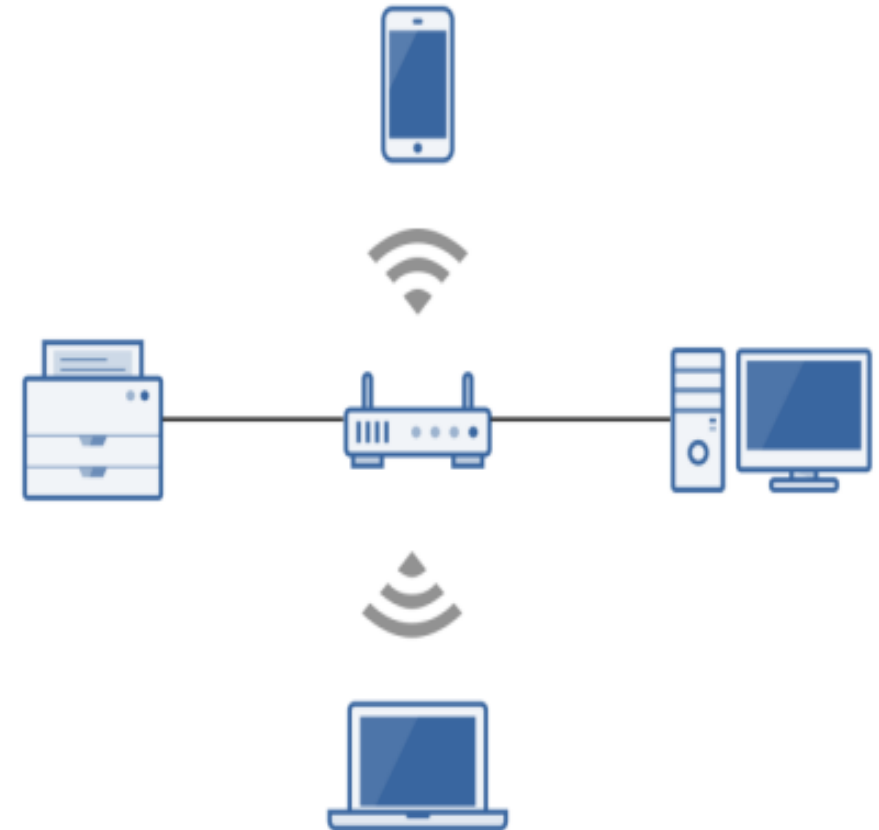
Personal Area Network (PAN)

- Ultra-small networks used for personal use to share data from one device to another.
- Can be wired (PAN) or wireless (WPAN):
 - USB
 - Bluetooth
 - NFC
 - ANT+
- Examples:
 - Smart Phone to Laptop
 - Smart Watch to Smart Phone
 - Smart Phone Hands-Free Car Calling
 - Heart Rate Monitor to Smart Phone



Local Area Network (LAN)

- A computer network within a small geographical area, such as a single room, building or group of buildings.
- Considered to be self-contained:
 - All devices are directly connected via cables and/or short-range wireless technology.
 - Doesn't require a leased telecommunications line from an Internet Service Provider (ISP).
- Examples:
 - Home Network
 - Small Business or Office Network



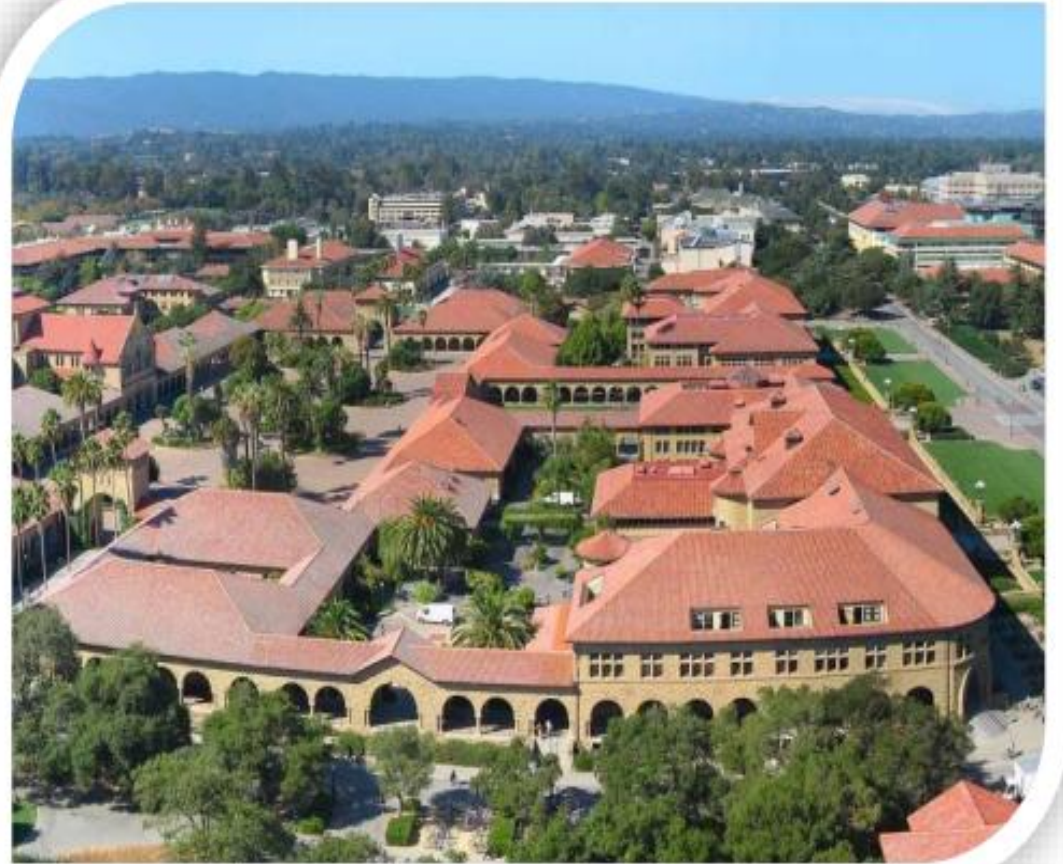
Wireless Local Area Network (WLAN)

- A LAN that's dependent on wireless connectivity or one that extends a traditional wired LAN to a wireless LAN.
- Most home networks are WLANs.



Campus Area Network (CAN)

- A computer network of multiple interconnected LANs in a limited geographical area, such as a corporate business park, government agency, or university campus.
- Typically owned or used by a single entity.



Metropolitan Area Network (MAN)

- A computer network that interconnects users with computer resources in a city.
- Larger than a campus area network, but smaller than a wide area network.



Wide Area Network (WAN)

- A computer network that extends over a large geographical distance, typically multiple cities and countries.
- WANs connect geographically distant LANs.
- Typically use leased telecommunications lines from ISPs.
- Examples:
 - The Internet
 - Corporate Offices in Different States



Some Basic Networking Rules

- The computers in a network must use the same procedures for sending and receiving data. We call these **communication protocols**.
- Data must be delivered uncorrupted. If it is corrupted, it's useless. (There are Exceptions)
- Computers in a network must be capable of determining the **origin** and **destination** of a piece of information, i.e., its **IP** and **Mac Address**.

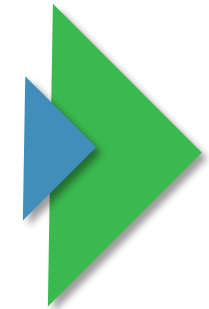
Computer Networking Protocols



- Computers communicate with each other with network protocols.
- Protocols are rules governing how machines exchange data and enable effective communication.
- **Some Everyday Examples**
 - When you call somebody, you pick up the phone, ensure there is a dial tone, and if there is, you dial the number.
 - When you drive your car, you obey the rules of the road.

Protocols Continued

- **Physical Protocols:** describe the medium (wiring), the connections (RJ-45 port), and the signal (voltage level on a wire).
- **Logical Protocols:** software controlling how and when data is sent and received to computers, supporting physical protocols.
- Computer networks depend on many different types of protocols in order to work properly.
- Example Common TCP/IP Suite of Protocols:
 - **Web Communication:** HTTP
 - **E-mail:** POP3, SMTP, IMAP
 - **File Transfers:** FTP



Network Topologies

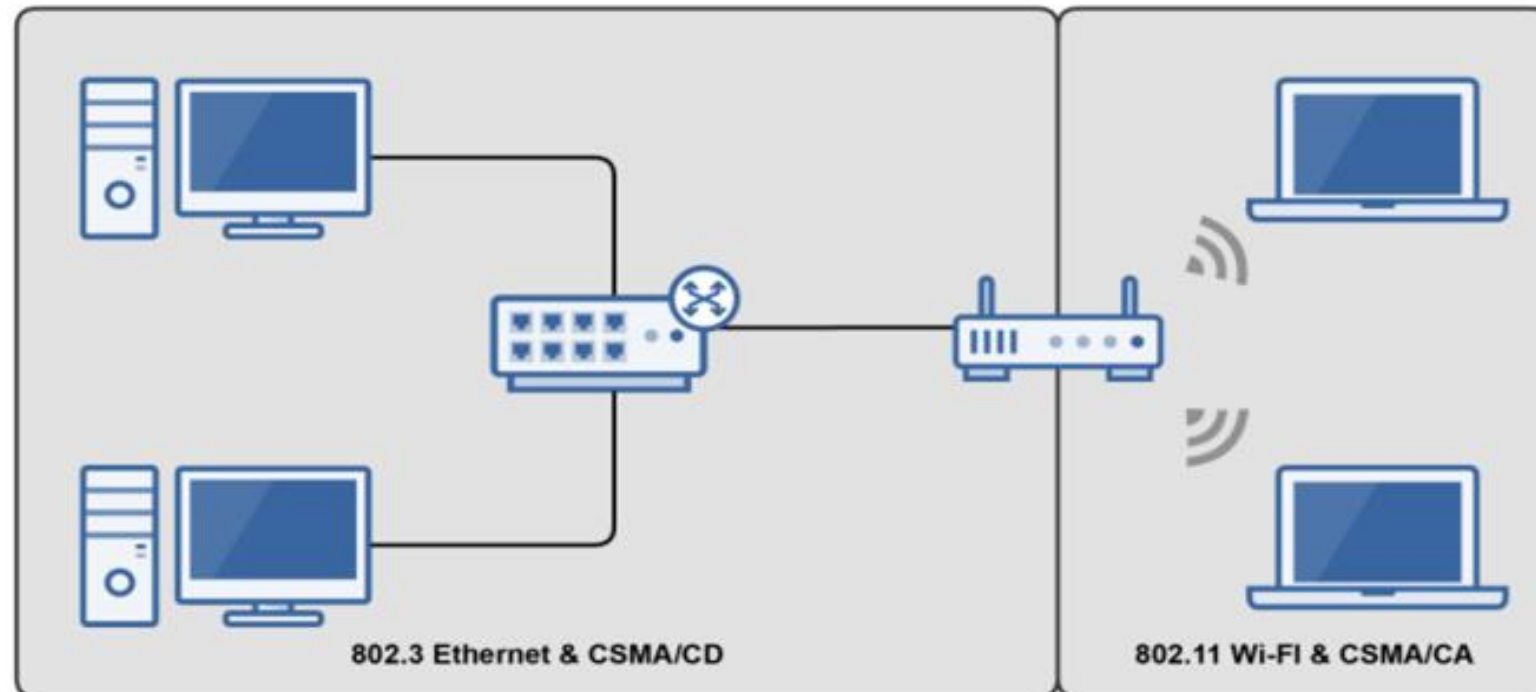
Introduction to Ethernet

- The most popular networking technology in the world!
- Refers to a family of standards that define the **physical** and **logical** aspects of the world's most popular type of LAN.
- The standard communications protocol for building a local area network (LAN).
- **Physical**
 - Cabling, Connectors, Equipment, etc.
- **Logical**
 - Network Access Method, i.e., Carrier Sense Multiple Access (CSMA)

Physical vs. Logical Topologies

Physical topologies describe the placement of network devices and how they are physically connected.

Logical topologies describe how data flows throughout a network.

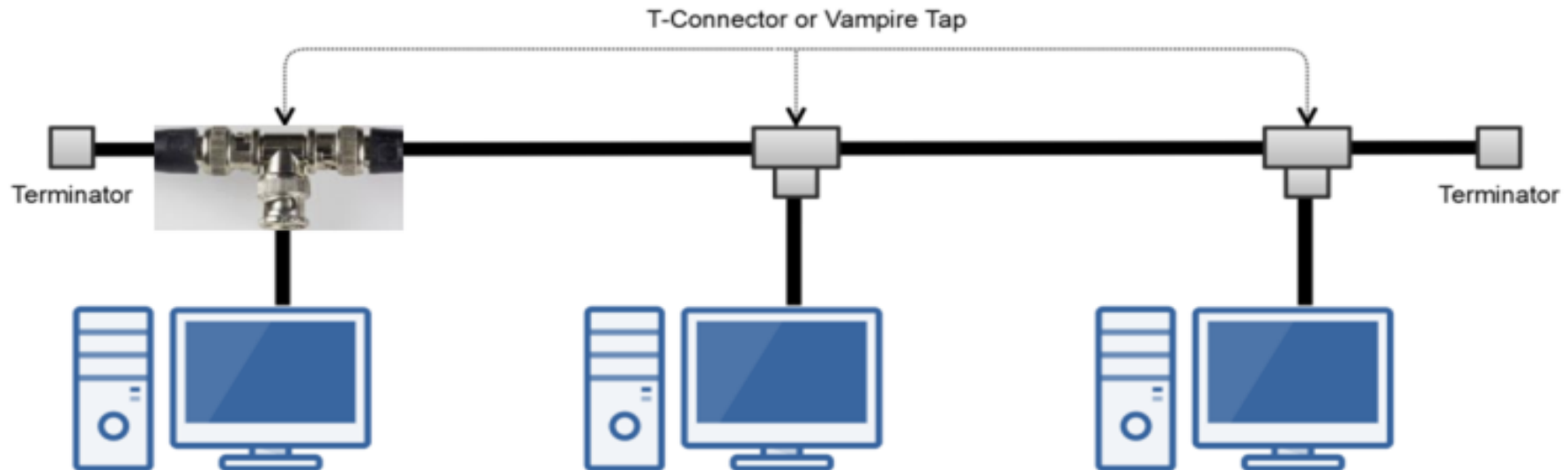


Wired Network Topologies

- Four Specific Topologies:
 - Bus
 - Ring
 - Star
 - Mesh

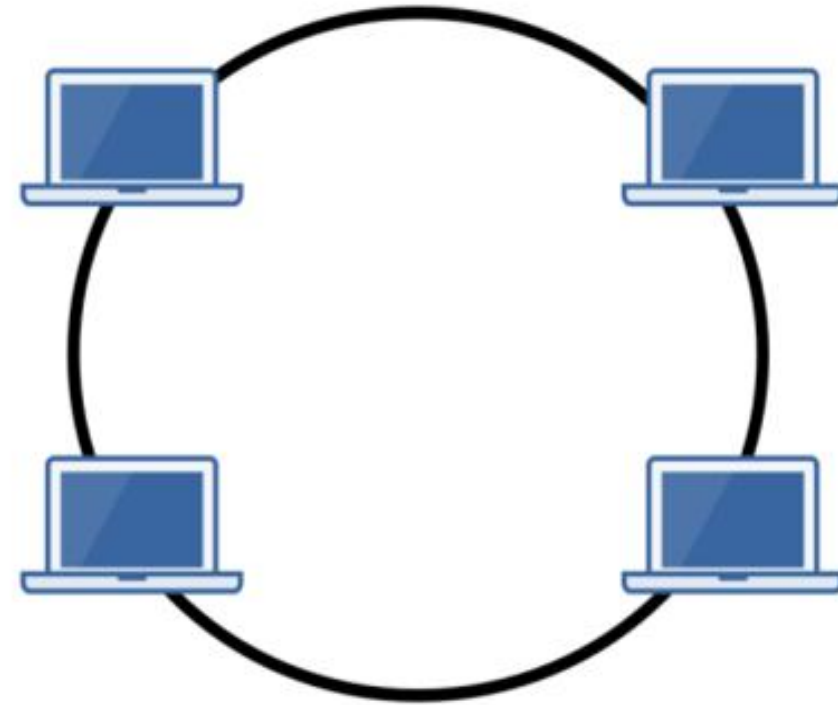
Bus Topology

- All devices are connected to a single coaxial network cable.
 - Devices are connected via a vampire tap or T-Connector.
 - Terminators are required at both ends of the cable to prevent signal bounce.
 - Antiquated technology.
- Only one end device can be active on the network at a time.
 - Data signals travel in both directions and are received by all devices on the network.
- A single break in the cable can take down the entire network.



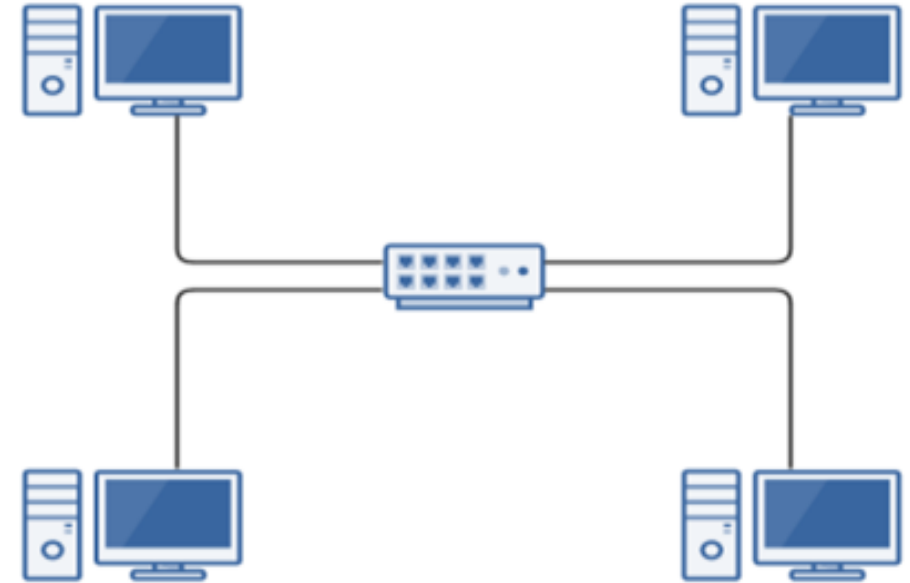
Ring Topology

- All devices are connected in a circular fashion.
- Each computer is connected to two other computers.
- Data travels from node-to-node with each computer handling data, either unidirectional or bidirectional.
- Each device (node) in the ring regenerates the signal, acting as a repeater.
- Failure of a single node can take down the entire network.
- Fiber Distributed Data Interface (FDDI) uses two counter-rotating ring topologies for redundancy.



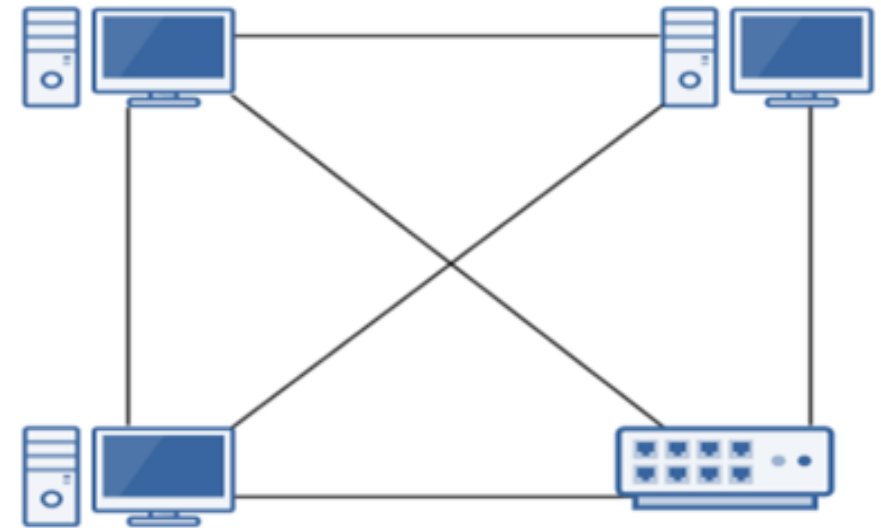
Star Topology

- All devices are connected to a central connecting device, which is usually a switch.
- Devices send data to the switch, which forwards it to the appropriate destination device.
- Popular topology in today's networks.
- Used in most large and small networks.
- Central device is a single point of failure.



Mesh Topology

- Each device is connected to every other device by separate cabling.
- Highly redundant and fault-tolerance.
- Expensive to install.
- Commonly used in Enterprise Networks & WANs.
- Two Types:
 - Partial Mesh
 - Full Mesh



Wireless Network Topologies

- Wireless networks utilize radio frequencies (RF) to communicate.
- Three Specific Topologies:
 - Ad hoc
 - Infrastructure
 - Mesh

Ad hoc

- Peer-to-peer (P2P) wireless network where no wireless access point (WAP) infrastructure exists.
- The devices communicate directly with one another.
- Personal area networks (PANs) are a common example of Ad hoc wireless networks.



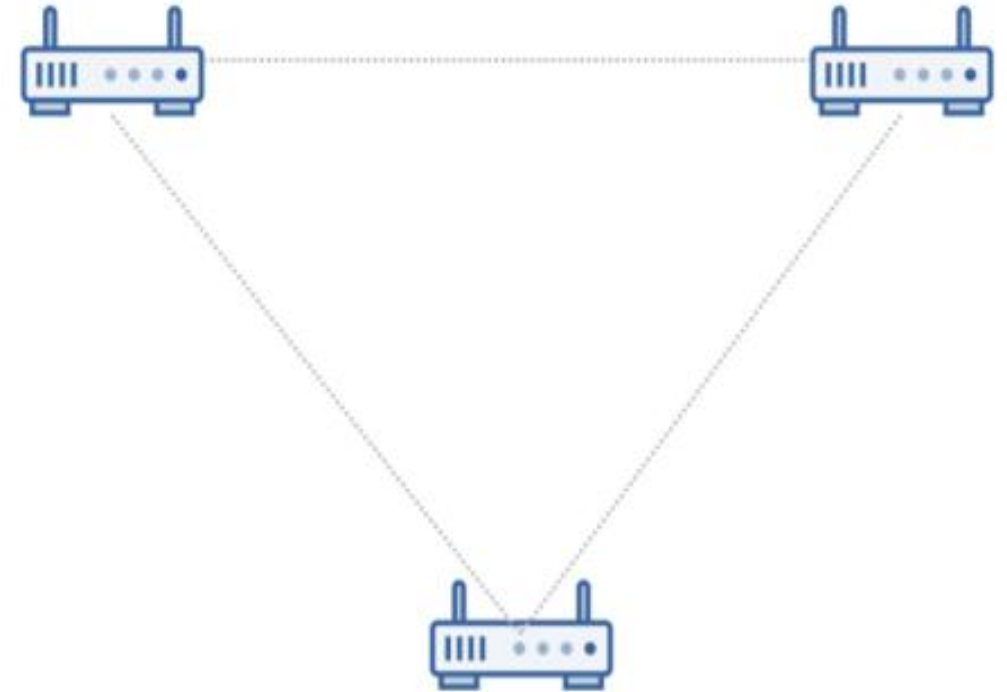
Infrastructure

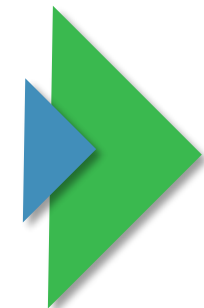
- Wireless network that uses a wireless access point (WAP) as its central connecting device.
- Infrastructure wireless networks (WLANs) are commonly used in homes and small offices.



Mesh

- Just like a wired mesh design, wireless mesh networks utilize several wireless access points (nodes) to create a robust wireless network that is:
 - Scalable
 - Self-Healing
 - Reliable (redundancy)
- Common in larger homes and businesses.





Types of Computer Network Architecture

Network Architecture

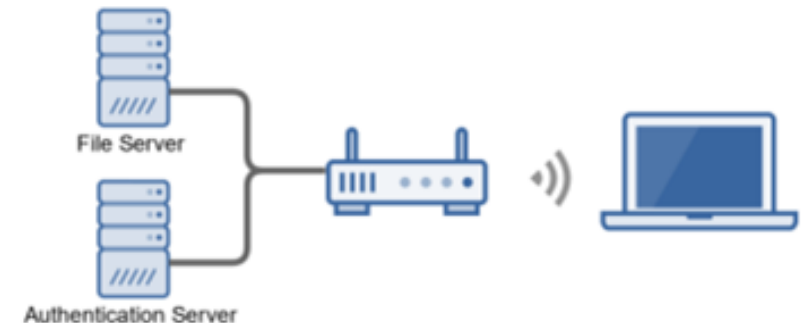
Peer-to-Peer

- All computers on the network are peers
 - No dedicated servers
 - There's no centralized control over shared resources
- Any device can share its resources as it pleases
- All computers can act as either a client or a server
- Easy to set-up, and common in homes and small businesses



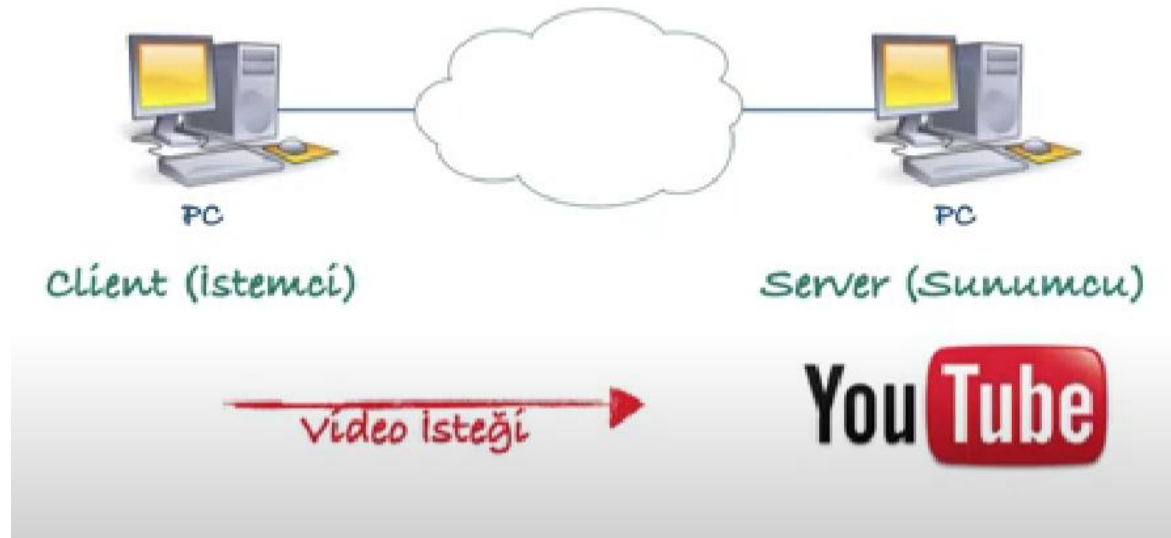
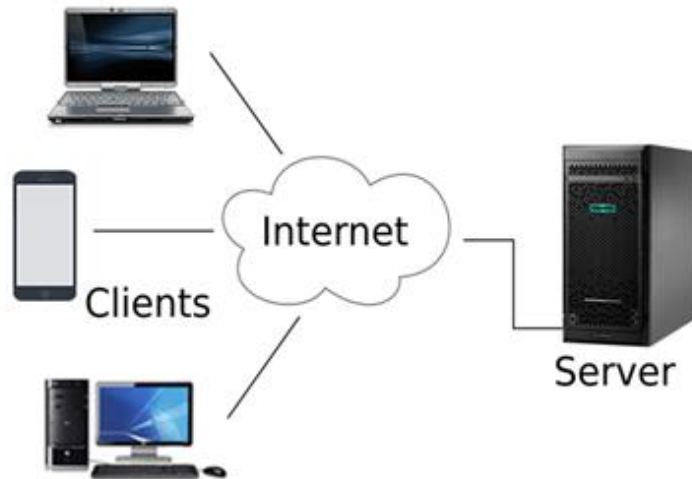
Client-Server

- The network is composed of client and servers
 - Servers provide resources
 - Clients receive resources
- Servers provide centralized control over network resources (files, printers, etc.)
- Centralizes user accounts, security, and access controls to simplify network administration
- More difficult to setup and requires an IT administrator



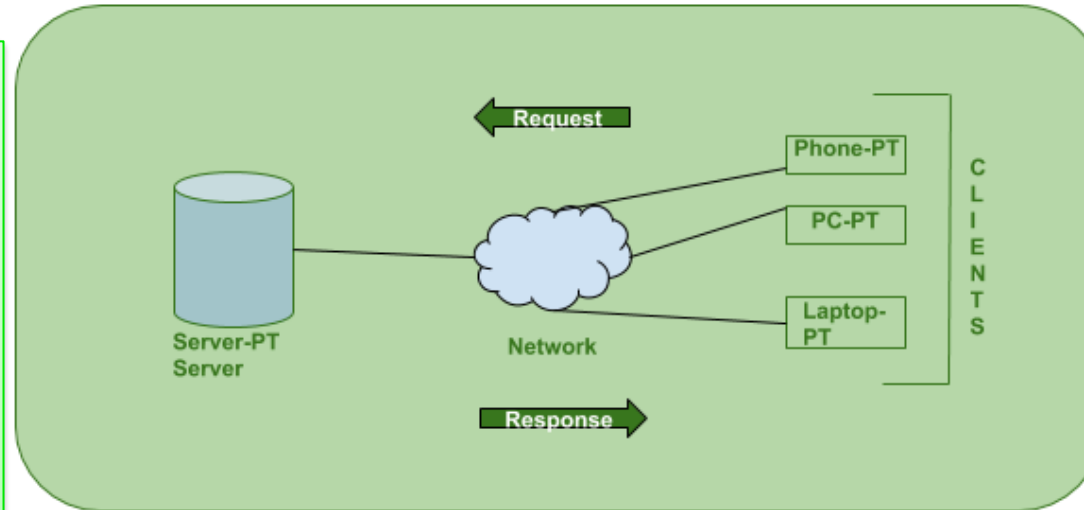
Client-Server Network Architecture

- A **Client-Server network architecture** is a structure where a **central server** provides services to **one or more client computers**.
- **Client computers** send requests to the **server**, and the **server responds** to these requests.
- Roles like **web servers**, **email servers**, and **database servers** are commonly used in this model.



Client-Server Network Architecture

- **Client:** When we talk about a *client*, we are referring to a person or organization that uses a particular service.
- In the digital world, a *Client* is a **computer (main computer)** that can request information or use specific services provided by *Servers*. So essentially, a client is a device (like a PC, phone, or laptop) that makes **requests** to a server.



Server: When we mention *servers*, we refer to entities that provide services or resources.

In the digital realm, a *Server* is a **remote computer** that provides access to data (information) or specific services.

In simple terms:

The **Client requests something**, and the **Server provides it** as long as the information exists in its database.



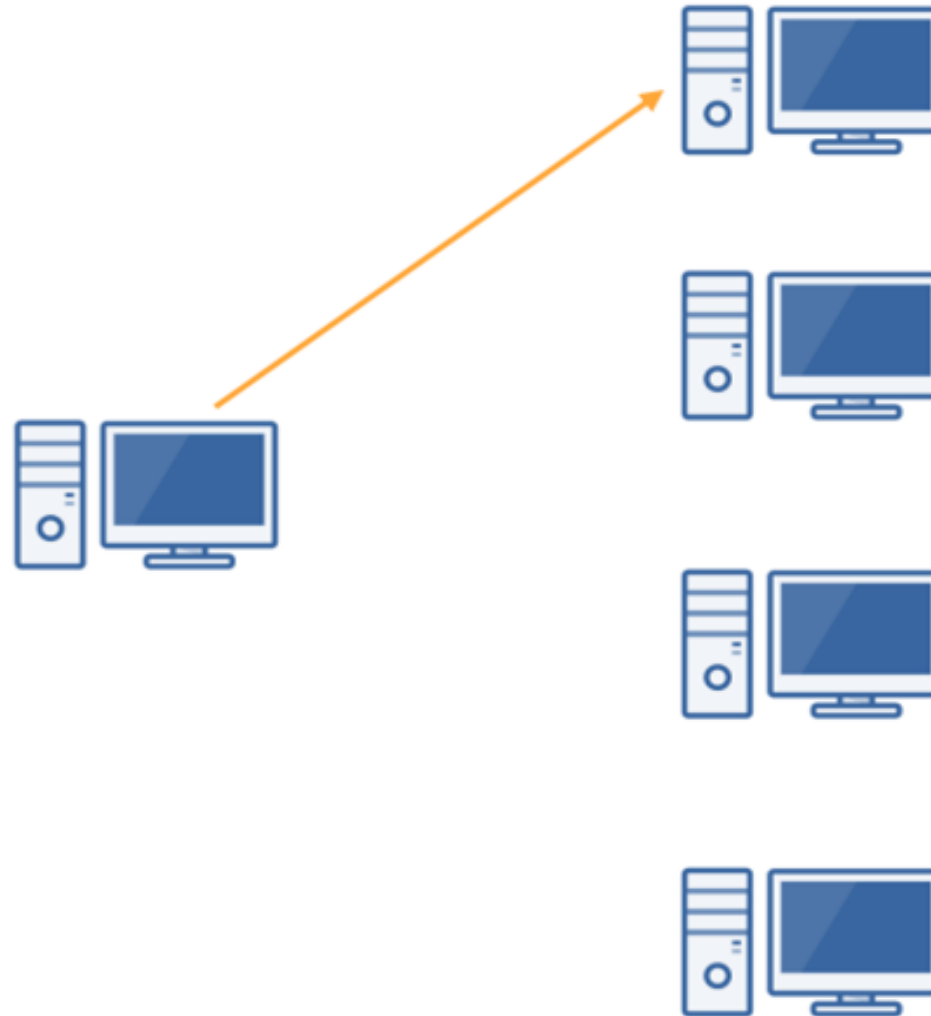
Types of Communication in The Network



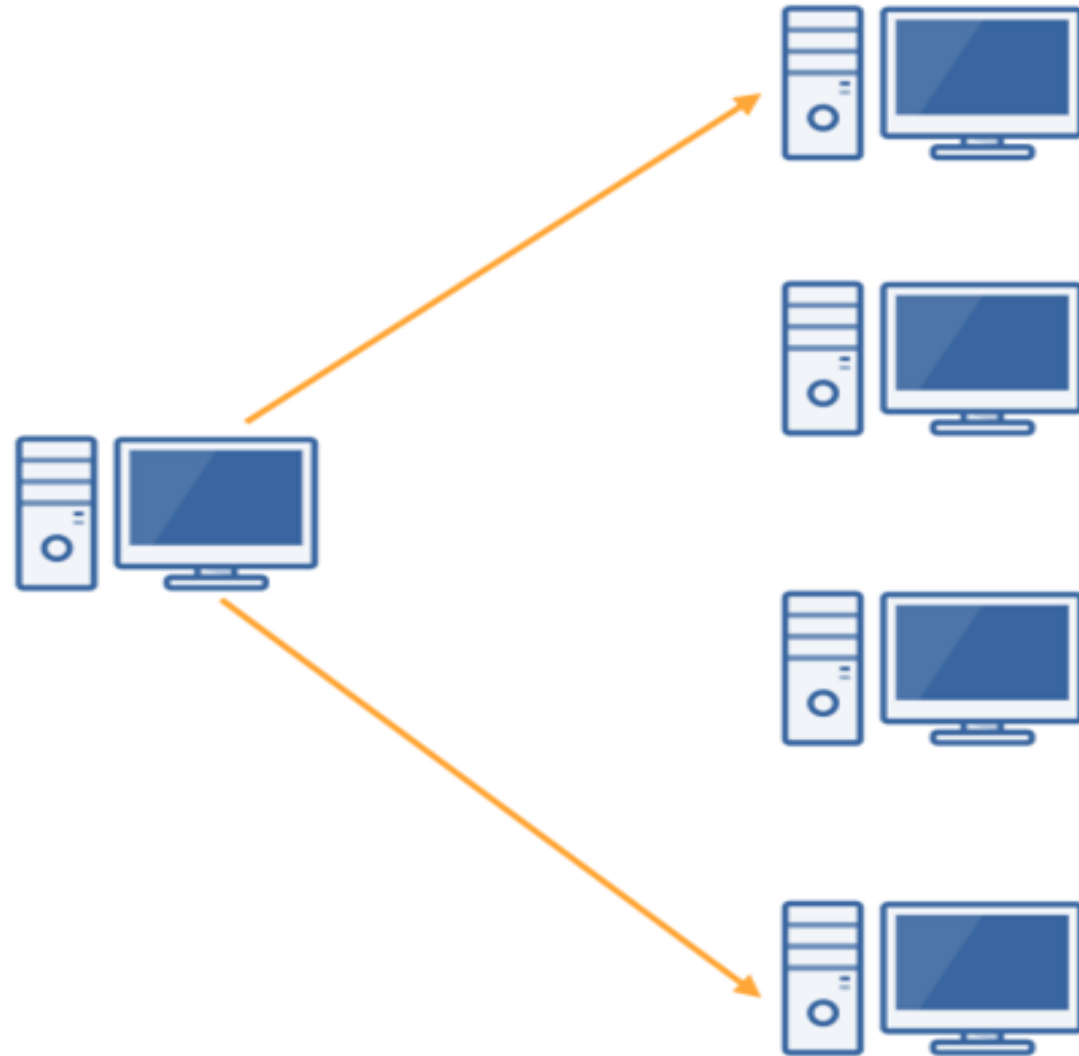
Network Transmission Types

- Unicast
- Multicast
- Broadcast

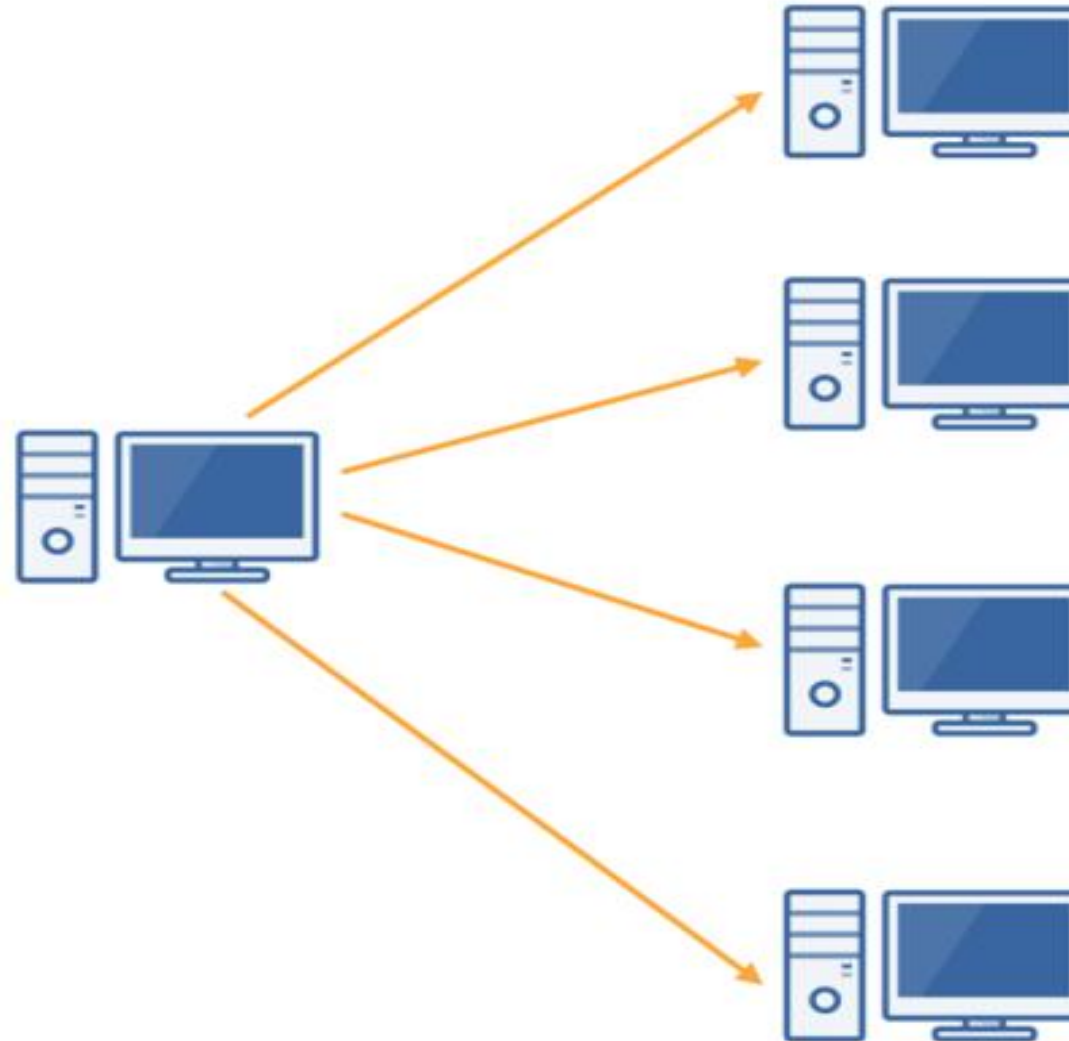
Unicast (One-to-One)



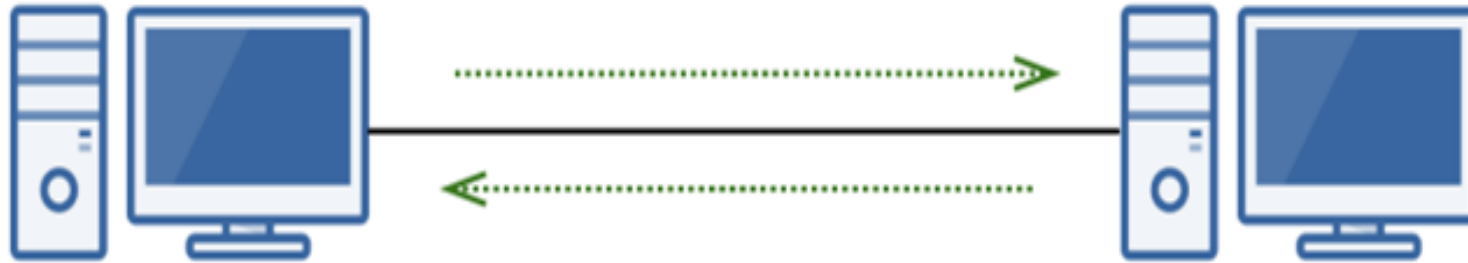
Multicast (One-to-Many)



Broadcast (One-to-All)



Half vs. Full Duplex Communication



- Network communication will occur in either full or half duplex mode:
 - **Half Duplex:** Can send and receive data, but not at the same time.
 - **Full Duplex:** Can send and receive data simultaneously.

Network Devices

MAC Addresses

Media Access Control (MAC)

- Physical address of the network adapter card
- **OSI Layer 2** (Data Link) Layer Address
- **TCP/IP Layer 1** (Network Interface) Layer Address



Six bytes (48 bits), Usually Represented Hexadecimal

- First three bytes (24 bits) are assigned by the IEEE to the manufacturer
 - Organizationally Unique Identifier (OUI) assigned by IEEE (ex: Dell or HP)
- Last three bytes (24 bits) are usually assigned sequentially:
 - Unique Numbers

00:21:70:6f:06:f2

00-21-70-6F-06-F2

$2^{24} = \sim 16.7$ Million Unique Addresses



IP Addresses

- An IP Address is a **logical** address used in order to **uniquely identify** a device on an IP network.
- It's a **Network Layer** address associated with routing.
 - **OSI Layer 3:** Network Layer
 - **TCP/IP Layer 2:** Internet Layer
- There are two versions:
 - **IP version 4 (IPv4)**
 - Example: 192.168.0.1
 - **IP version 6 (IPv6)**
 - Example: 2001:DB8:85A3:0:0:8A2E:370:7334
- We'll be discussing both versions in this course.

Comparing IP and MAC Addresses

IP Addresses

- Network (OSI Layer 3) Addresses
- Logical Addresses
- Assigned in Operating System
- Allows network-to-network communication via routers
- WAN communication

MAC Addresses

- Data Link (OSI Layer 2) Addresses
- Physical Addresses
- Physically burned on NIC
- Allows internetwork communication via hubs, switches, and routers
- Local LAN communication

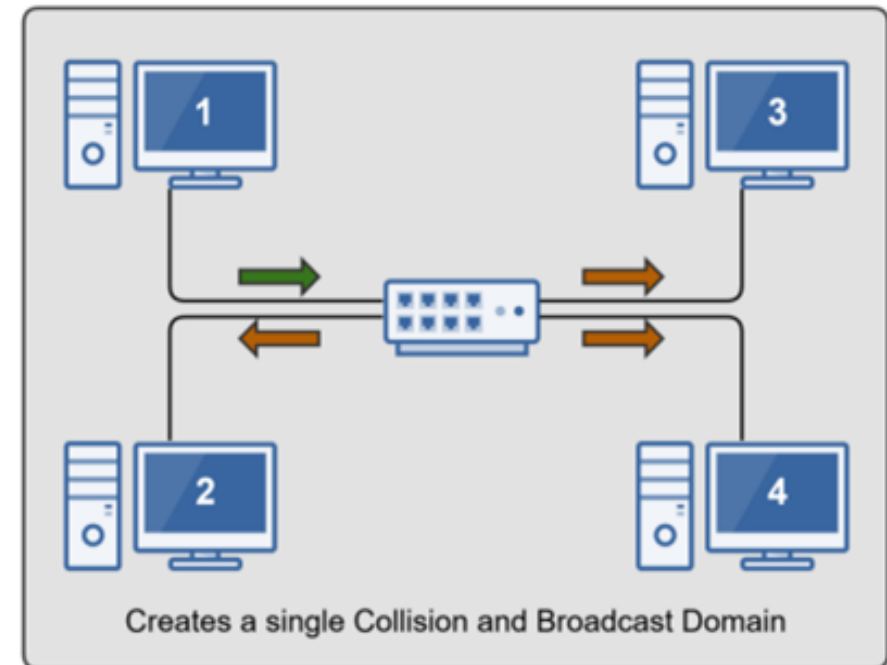
Network Interface Card (NIC)

- The network adapter installed on your network device.
- Provides the physical and electrical, light or radio frequency connections to the network media.
- It can either be an expansion card, USB devices or built directly into the motherboard.



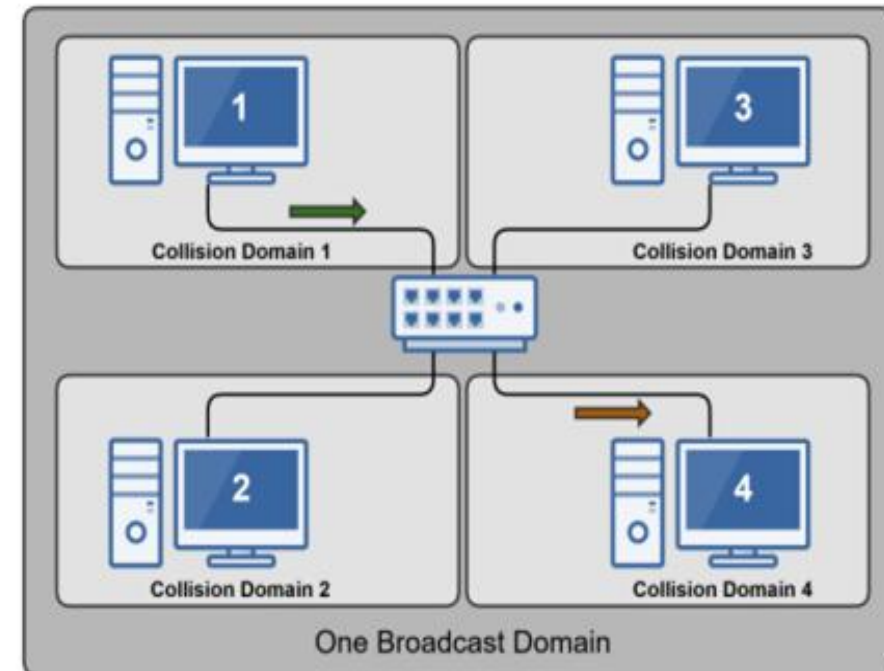
Hubs

- Used to Connect Devices Together Within a Network
- Used in Early Networks; Replaced by Switches
- “Multi-Port Repeater”
 - Traffic goes in one port and is repeated (broadcasted) out every other port
 - OSI Layer 1 Device
 - Dumb Network Device
 - Causes increased network collision errors
- Much Less Efficient than a Switch
- Legacy Equipment No Longer Used



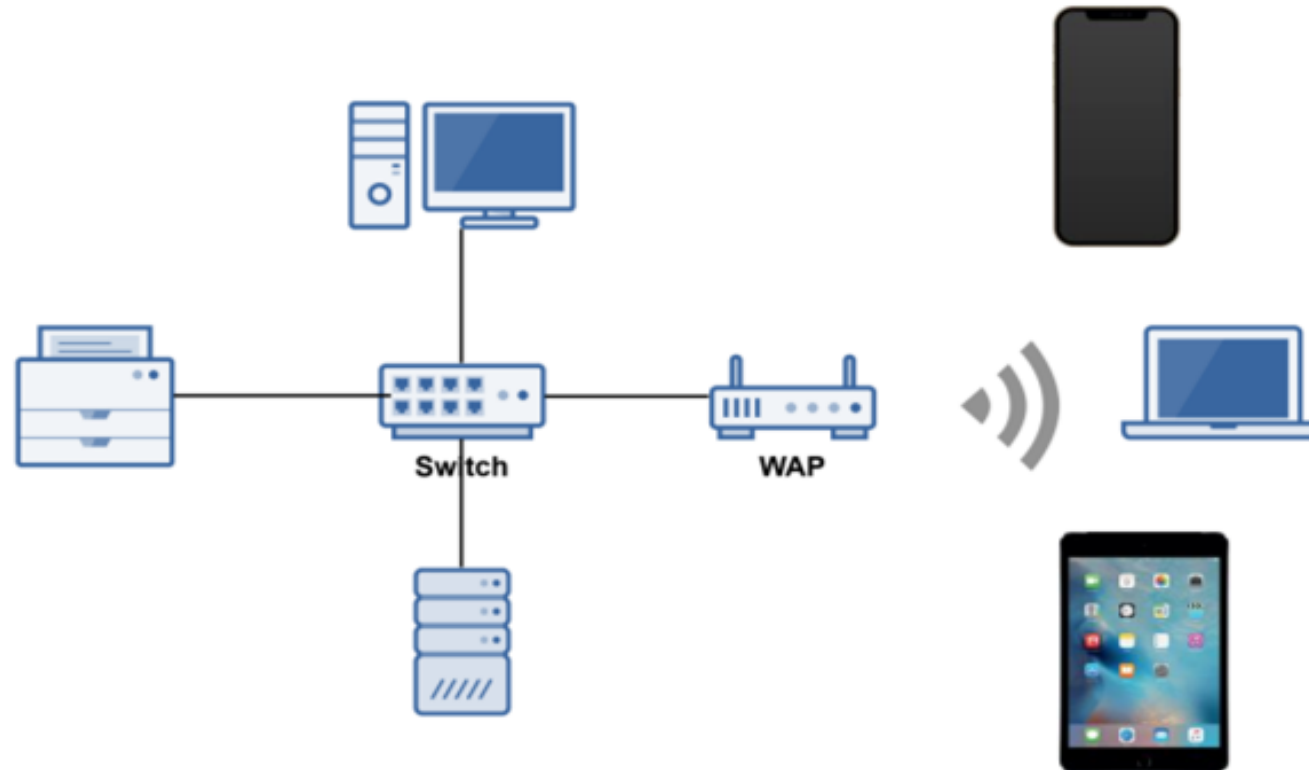
Switches

- Connects Devices Together Just Like a Hub
- Intelligent Network Device (OSI Layer 2)
- Memorizes the **MAC Address** of Each Device Connected to It via a **MAC Address Table**, sometimes called a **Content Addressable Memory (CAM) Table**
- Pays attention to *Source* and *Destination* **MAC addresses** during Communication Process
- Use Application-Specific Integrated Circuitry (**ASIC**), which makes them Extremely Fast
- Breaks up Collision Domains
 - Traffic Goes in One Port and Is Repeated out to Only Destination Port
 - Designed for High Bandwidth
 - Standard in Today's Network Infrastructure



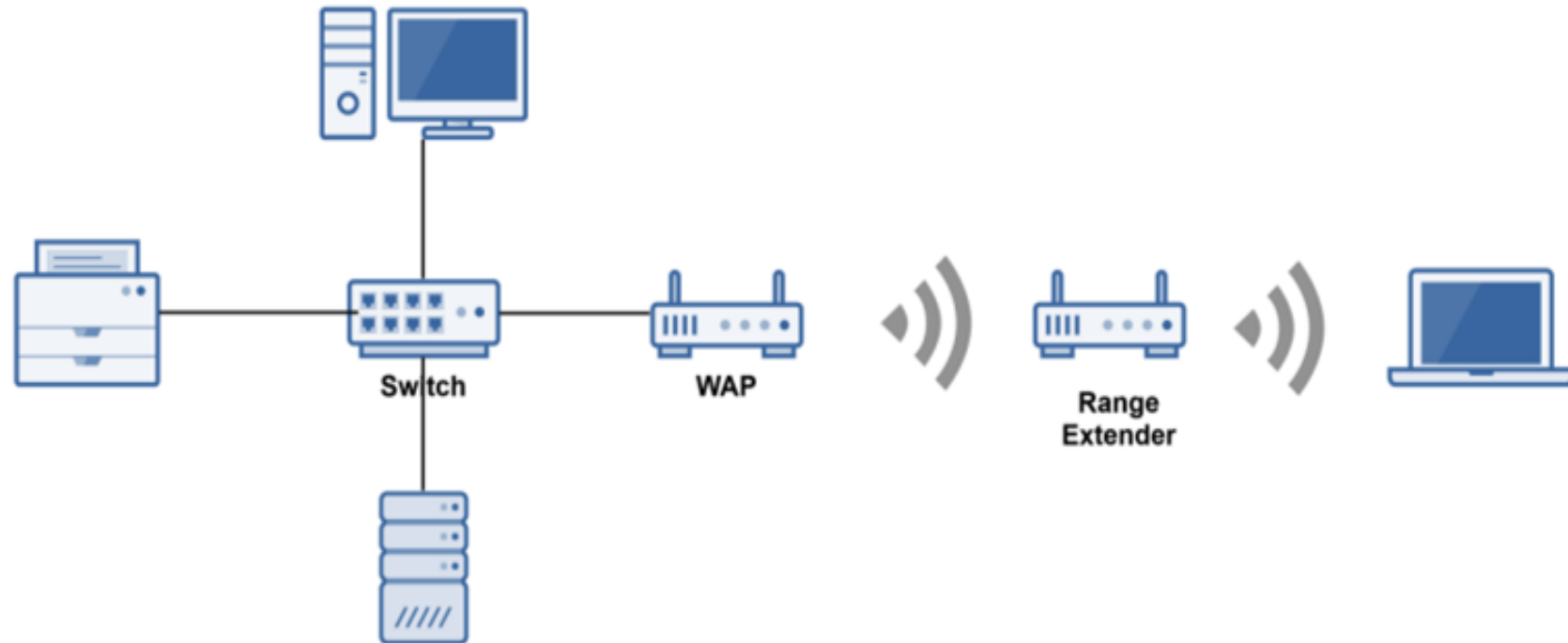
Wireless Access Point (WAP)

- A wireless access point (WAP) is a bridge that extends the wired network to the wireless network.
- Just like a switch, it's a Data Link Layer 2 device.
- **Note:** A WAP is not a router.



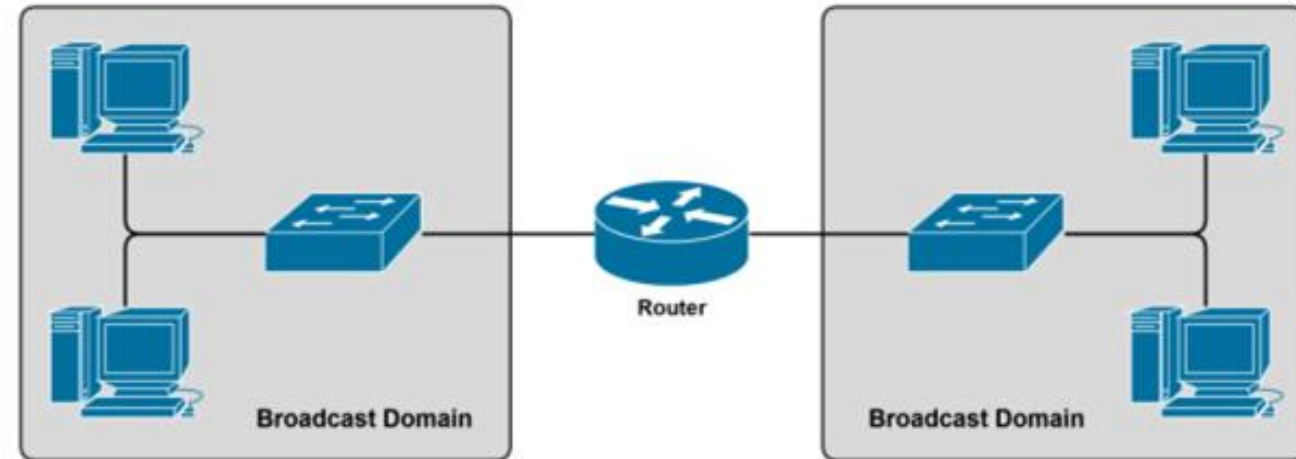
Wireless Ranger Extender

- Extends the range of a wireless network by acting as a wireless repeater.
- Rebroadcasts radio frequencies from the wireless network it is associated with



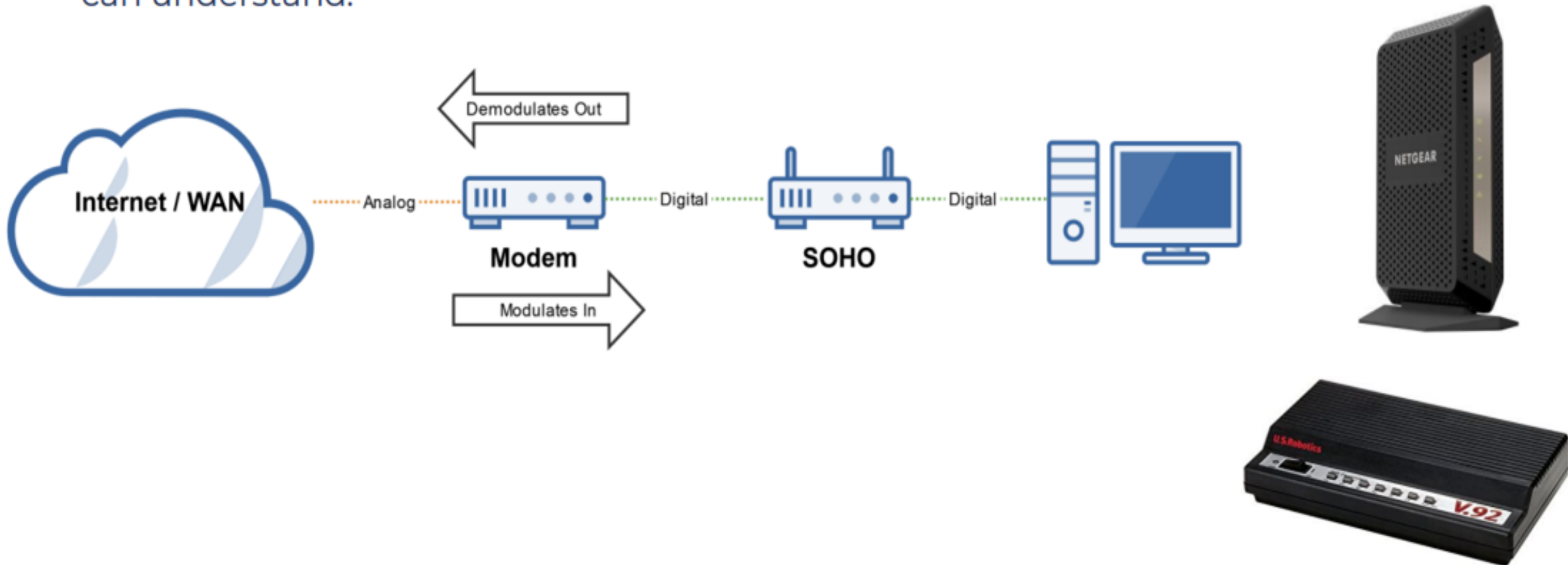
Routers

- Used to Connect Different Networks Together
- Routes Traffic Between Networks using **IP Addresses**
- Uses Intelligent Decisions (Routing Protocols) to Find the Best Way to Get a Packet of Information from One Network to Another.
- Break Up Broadcast Domains
- **OSI Layer 3 Device**
 - Layer 3 = Router
 - Layer 2 = Switch
 - Layer 1 = Hub



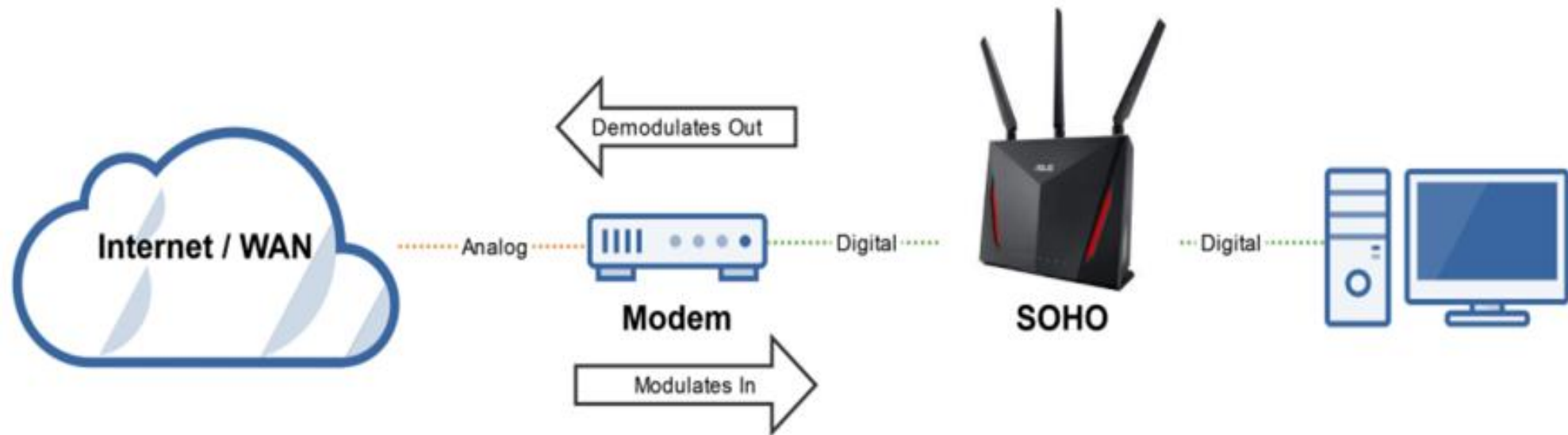
Modems (Modulators/Demodulators)

- Modems modulate one signal to another, such as analog to digital.
- For example, modulating a telephone analog signal into a digital signal that a router can understand.



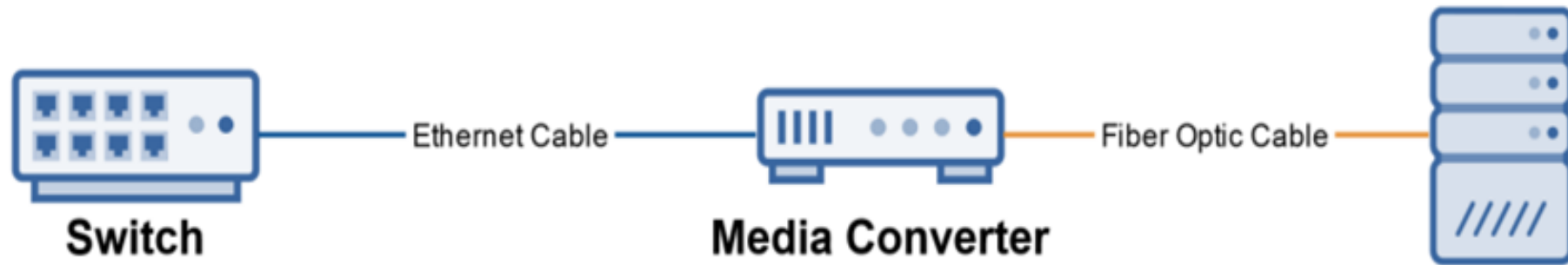
Small Office Home Office (SOHO) Device

- All-In-One Wireless Router with Expanded Capabilities:
 - Router, Wireless Access Point, Firewall, Switch, DHCP Server, NAT Device, File Server, etc.



Media Converters

- Like its name implies, it converts one media type to another.
- **Layer 1 Device:** Performs physical layer signal conversion.
- Ethernet to fiber optic media converters are commonly used.



Firewalls

- Firewalls are the foundation of a defense-in-depth network security strategy.
- They protect your network from malicious activity on the Internet.
- Prevent unwanted network traffic on different networks from accessing your network.
- Firewalls do this by filtering data packets that go through them.
- They can be a standalone network device or software on a computer system, meaning **network-based (hardware)** or **host-based (software)**.



Types of Firewalls

Packet Filtering Firewalls

- 1st Generation & Most Basic
- Basic Filtering Rules

Circuit-Level Firewalls

- 2nd Generation
- Monitors Valid/Invalid TCP Sessions

Application Layer 7 (NGFW) Firewalls

- 3rd Generation
- Much more Advanced; Covered Later in Course

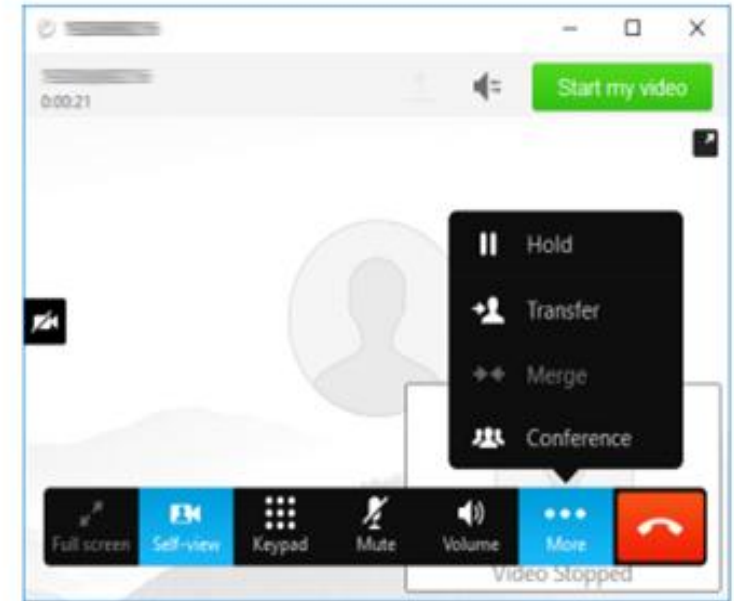
DHCP Server

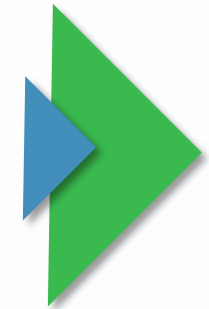
- Dynamic Host Configuration Protocol (DHCP) Server
- Automatically Assigns IP Addresses to Hosts
- Makes Administering a Network Much Easier
- An Alternative is Static IP addressing
- We'll Talk More About DHCP Later in the Course



Voice over IP (VoIP) Endpoints

- Most phone systems run over IP networks via dedicated protocols, such as the Session Initiation Protocol (SIP), both in-home and office environments.
- VoIP endpoint devices are hardware devices (phones) or software, such as Cisco Jabber, that allow you to make phone calls.





Network Terminologies

Network Terminologies

- **Nodes:** Devices connected to a network. These can include computers, servers, printers, routers, switches, and other devices.
- **Host:** A main computer or device on a network. A host can perform data transactions or operations on the network.
- **Router:** A device that directs data packets to their correct destination across computer networks. It provides internet access and manages network traffic.
- **Switch:** A device that connects devices within a network and forwards data only to the specific device it is intended for. Common in Ethernet networks for efficient data delivery.
- **Firewall:** A security device that monitors and controls incoming and outgoing network traffic. It blocks unwanted or harmful traffic and is essential for network security.
- **IP Address:** A unique identifier assigned to a device on a network. It can be in IPv4 or IPv6 format (e.g., 192.168.1.1 for IPv4).
- **DNS (Domain Name System):** A system that translates domain names (like www.example.com) into IP addresses. Essential for accessing websites by name instead of number.
- **OSI (Open Systems Interconnection):** A conceptual framework developed by ISO and ITU-T to standardize network communication functions. It ensures interoperability between different vendors and devices through 7 structured layers (like Physical, Data Link, Network, Transport, etc.).

Network Terminologies

- LAN (Local Area Network):**

A network that connects devices within a limited geographical area (e.g., an office or home).

- WAN (Wide Area Network):**

A network that spans a larger geographical area (e.g., between cities or countries). The internet is a type of WAN.

- Ethernet:**

A widely used communication technology for wired network connections. Ethernet cables connect devices to the network.

- Gateway:**

A device that allows data to flow between different networks or communication protocols. It routes data between two distinct networks.

- Packet:**

A small unit of data sent over a network. In computer communication, data is broken into packets for transmission.

- Bandwidth:**

The capacity of a network connection to transfer data. It's usually measured in megabits or gigabits per second (Mbps or Gbps).

- Protocol:**

A set of rules that determine how devices communicate and exchange data on a network. For example, TCP/IP is a widely used network protocol.

Network Terminologies

- ARP (Address Resolution Protocol):**

A protocol defined by RFC 826 that finds the MAC address from an IP address.

- MAC Address (Media Access Control Address):**

A unique identifier assigned to a device on a network. Every network interface card (NIC) or network device has a MAC address.

- www (World Wide Web):**

The abbreviation for "World Wide Web," a system that contains documents (web pages) available on the internet.

- URL (Uniform Resource Locator):**

A unique address system used to identify resources (web pages, files, images, videos, etc.) on the internet.

- DHCP (Dynamic Host Configuration Protocol):**

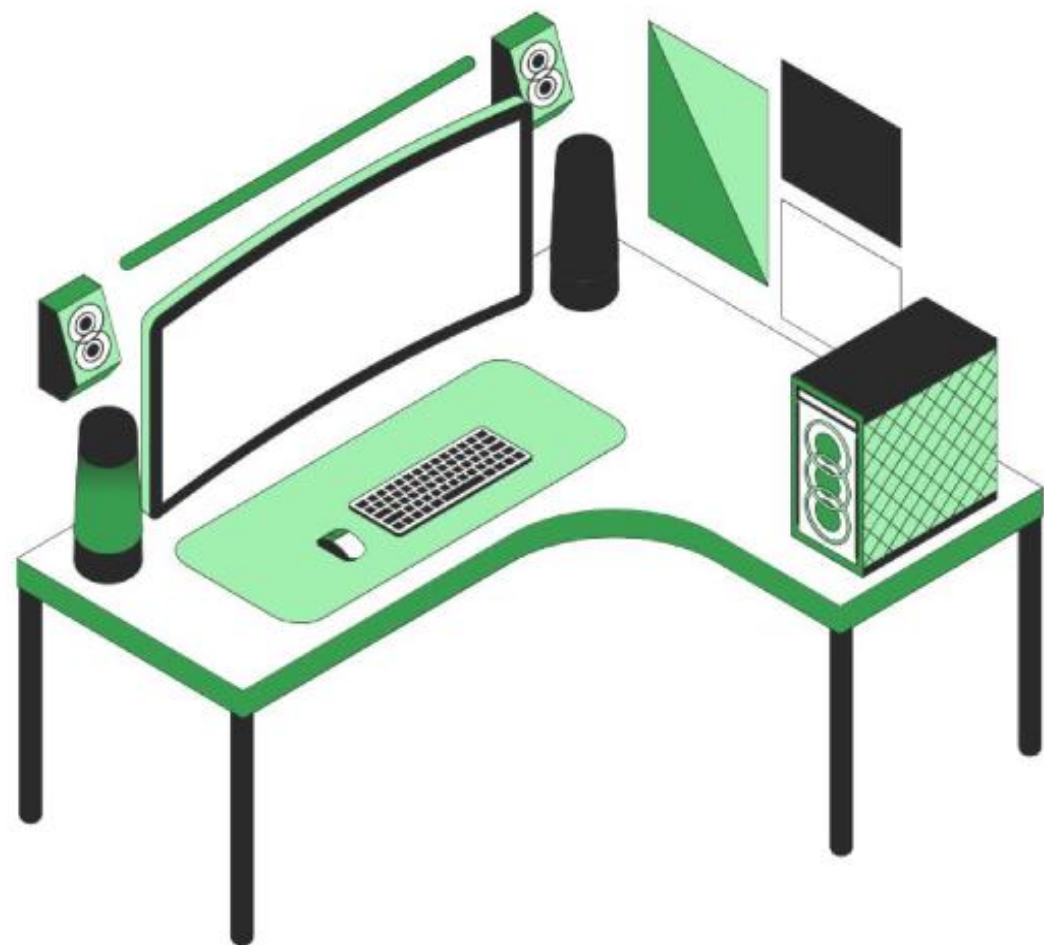
A communication protocol that automatically assigns IP addresses and manages network configuration for devices on the network.

- TCP (Transmission Control Protocol):**

A communication protocol designed to ensure reliable and orderly transmission of data.

- UDP (User Datagram Protocol):**

A faster but less reliable communication protocol compared to TCP.



Do you have any questions?

Send it to us! We hope you learned something new.